



Recitation 4

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1. Homework

2. Review

3. Lecture

4. Recitation



1. **Homework**

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1. A source transmits a message (a string of symbols) through a noisy communication channel. Each symbol is 0 or 1 with probability p and $1 - p$, respectively, and is received incorrectly with probability ϵ_0 and ϵ_1 , respectively. Errors in different symbol transmissions are independent.

(e) Suppose that the scheme of part (c) is used. What is the probability that a symbol was 0 given that the received string is 101?

1. A source transmits a message (a string of symbols) through a noisy communication channel. Each symbol is 0 or 1 with probability p and $1 - p$, respectively, and is received incorrectly with probability ϵ_0 and ϵ_1 , respectively. Errors in different symbol transmissions are independent.

(e) Suppose that the scheme of part (c) is used. What is the probability that a symbol was 0 given that the received string is 101?

Compute $P(0|101)$ or $P(101|0)$?

3. Show that the conditional independence of A and B given C neither implies, nor is implied by, the independence of A and B . For which events C is it the case that, for all A and B , the events A and B are independent if and only if they are conditionally independent given C ?



3. Show that the conditional independence of A and B given C neither implies, nor is implied by, the independence of A and B . For which events C is it the case that, for all A and B , the events A and B are independent if and only if they are conditionally independent given C ?

For (2), some students give the answer "when C is independent of A and B ." Notice that the property of C should hold for all A and B instead of specific one.

4. Jane has three children, each of which is equally likely to be a boy or a girl independently of the others. Define the events:

$A = \{\text{all the children are of the same sex}\},$

$B = \{\text{there is at most one boy}\},$

$C = \{\text{the family includes a boy and a girl}\}.$

(c) Do these results hold if boys and girls are not equally likely?

4. Jane has three children, each of which is equally likely to be a boy or a girl independently of the others. Define the events:

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(c) Do these results hold if boys and girls are not equally likely?

Compute $P(A \cap B)$ and $P(A) \cdot P(B)$. Although the above two do not have the same expression, you can still find the equation $P(A \cap B) = P(A) \cdot P(B)$ in special cases.

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2. Review

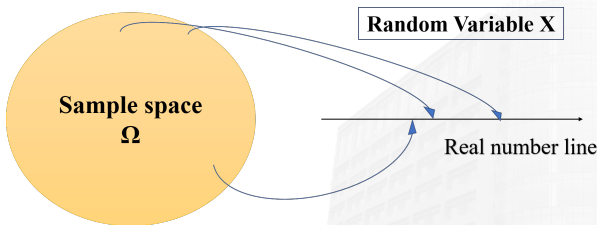
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- Random variables: A **function** from the sample space Ω to the real numbers
- PMF (for discrete random variables):

$$p_X(x) = \mathbf{P}(X = x) = \mathbf{P}\{\omega : X(\omega) = x\}$$



- Expectation

$$\mathbf{E}[X] = \sum_x x p_X(x)$$

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x)$$

$$\mathbf{E}[\alpha X + \beta] = \alpha \mathbf{E}[X] + \beta$$

- Variance

$$\begin{aligned}\text{var}(X) &= \mathbf{E}[(X - \mathbf{E}[X])^2] \\ &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2\end{aligned}$$

$$\text{var}(\alpha X + \beta) = \alpha^2 \text{var}(X)$$

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- **Bernoulli RV X with PMF**

$$p_X(k) = \begin{cases} p, & \text{if } k = 1 \\ 1 - p, & \text{if } k = 0 \end{cases}$$

- **Mean of RV X**

$$\mathbf{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p$$

- **Second moment of RV X**

$$\mathbf{E}[X^2] = 1^2 \cdot p + 0 \cdot (1 - p) = p$$

- **Variance of RV X**

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = p - p^2 = p(1 - p)$$

- Discrete Uniform RV X with PMF

$$p_X(k) = \begin{cases} \frac{1}{b-a+1}, & \text{if } k = a, a+1, \dots, b \\ 0, & \text{otherwise} \end{cases}$$

- Mean of RV X

$$\mathbf{E}[X] = \frac{a+b}{2}$$

- Variance of RV X

$$\text{var}(X) = \frac{(b-a+1)^2 - 1}{12} = \frac{(b-a)(b-a+2)}{12}$$

- **Poisson RV** X with PMF

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

- Mean of **RV** X

$$\begin{aligned} \mathbf{E}[X] &= \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} \quad (\text{the } k=0 \text{ term is } 0) \\ &= \lambda \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda \sum_{m=0}^{\infty} e^{-\lambda} \frac{\lambda^m}{m!} \quad (\text{let } m = k-1) \\ &= \lambda \end{aligned}$$

- **Variance of RV:** $\text{var}(X) = \lambda$

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1. Some proofs.

- (a) Derive the expected value rule for functions of random variables

$$\mathbf{E}[g(X)] = \sum_x g(x)p_X(x)$$

- (b) Derive the property for the mean and variance of a linear function of a random variable $Y = aX + b$

$$\mathbf{E}[Y] = a\mathbf{E}[X] + b, \quad \text{var}(Y) = a^2\text{var}(X)$$

- (c) Derive $\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$

1. Some proofs.

(a) Derive the expected value rule for functions of random variables

$$\mathbf{E}[g(X)] = \sum_x g(x)p_X(x)$$

Let $Y = g(X)$ and we have $p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$. Thus,

$$\begin{aligned}\mathbf{E}[g(X)] &= \mathbf{E}[Y] = \sum_y y \sum_{\{x|g(x)=y\}} p_X(x) \\ &= \sum_y \sum_{\{x|g(x)=y\}} yp_X(x) \\ &= \sum_y \sum_{\{x|g(x)=y\}} g(x)p_X(x) \\ &= \sum_x g(x)p_X(x).\end{aligned}$$

1. Some proofs.

(b) Derive the property for the mean and variance of a linear function of a random variable $Y = aX + b$

$$\mathbf{E}[Y] = a\mathbf{E}[X] + b, \quad \text{var}(Y) = a^2\text{var}(X)$$

$$\mathbf{E}[Y] = a \sum_x xp_X(x) + b \sum_x p_X(x) = a\mathbf{E}[X] + b.$$

$$\begin{aligned} \text{var}(Y) &= \sum_x (ax + b - a\mathbf{E}[X] - b)^2 p_X(x) \\ &= a^2 \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= a^2 \text{var}(X). \end{aligned}$$

1. Some proofs.

(c) Derive $\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$

$$\begin{aligned}\text{var}(X) &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= \sum_x (x^2 - 2x\mathbf{E}[X] + (\mathbf{E}[X])^2) p_X(x) \\ &= \mathbf{E}[X^2] - 2(\mathbf{E}[X])^2 + (\mathbf{E}[X])^2 \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2.\end{aligned}$$

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

- (a) Calculate and sketch the PMF of X .
- (b) What is the probability of scoring no hits?
- (c) What is the probability of scoring more hits than misses?

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

(a) Calculate and sketch the PMF of X .

X is a Binomial random variable with $n = 10, p = 0.2$. Therefore,

$$p_X(k) = \binom{10}{k} 0.2^k 0.8^{10-k}, \quad \text{for } k = 0, \dots, 10$$

and $p_X(k) = 0$ otherwise.

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

(b) What is the probability of scoring no hits?

$$\mathbf{P}(\text{No hits}) = p_X(0) = 0.8^{10} = 0.1074.$$

(c) What is the probability of scoring more hits than misses?

$$\mathbf{P}(\text{More hits than misses}) = \sum_{k=6}^{10} p_X(k) = 0.0064.$$

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

- (d) Find the expectation and the variance of X .
- (e) Suppose the marksman has to pay \$3 to enter the shooting range and he gets \$2 dollars for each hit. Let Y be his profit. Find the expectation and the variance of Y .
- (f) Now let's assume that the marksman enters the shooting range for free and gets the number of dollars that is equal to the square of the number of hits. Let Z be his profit. Find the expectation of Z .

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

(d) Find the expectation and the variance of X .

Since X is Binomial random variable,

$$\mathbf{E}[X] = 10 \cdot 0.2 = 2, \quad \text{var}(X) = 10 \cdot 0.2 \cdot 0.8 = 1.6.$$

(e) Suppose the marksman has to pay \$3 to enter the shooting range and he gets \$2 dollars for each hit. Let Y be his profit. Find the expectation and the variance of Y .

$Y = 2X - 3$, and therefore

$$\mathbf{E}[Y] = 2\mathbf{E}[X] - 3 = 1, \quad \text{var}(Y) = 4 \cdot \text{var}(X) = 6.4.$$

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

- (f) Now let's assume that the marksman enters the shooting range for free and gets the number of dollars that is equal to the square of the number of hits. Let Z be his profit. Find the expectation of Z .

$Z = X^2$, and therefore

$$\mathbf{E}[Z] = \mathbf{E}[X^2] = (\mathbf{E}[X])^2 + \text{var}(X) = 5.6$$

3. 4 buses carrying 148 job-seeking MIT students arrive at a job convention. The buses carry 40, 33, 25, and 50 students, respectively. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying this randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on his bus.

- (a) Which of $\mathbf{E}[X]$ or $\mathbf{E}[Y]$ do you think is larger? Give your reasoning in words.
- (b) Compute $\mathbf{E}[X]$ and $\mathbf{E}[Y]$.

3. 4 buses carrying 148 job-seeking MIT students arrive at a job convention. The buses carry 40, 33, 25, and 50 students, respectively. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying this randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on his bus.

- (a) Which of $\mathbf{E}[X]$ or $\mathbf{E}[Y]$ do you think is larger? Give your reasoning in words.

We expect $\mathbf{E}[X]$ to be higher than $\mathbf{E}[Y]$ since if we choose the student, we are more likely to pick a bus with more students.

(b) Compute $\mathbf{E}[X]$ and $\mathbf{E}[Y]$.

We first compute the PMF of each random variable and then compute their expectations.

$$p_X(x) = \begin{cases} 40/148, & x = 40, \\ 33/148, & x = 33, \\ 25/148, & x = 25, \\ 50/148, & x = 50, \\ 0, & \text{otherwise.} \end{cases}$$

and $\mathbf{E}[X] = 39.28$.

$\mathbf{E}[Y] = \frac{1}{4}(40 + 33 + 25 + 50) = 37$.

4. **St. Petersburg paradox.** You toss independently a fair coin and you count the number of tosses until the first tail appears. If this number is n , you receive 2^n dollars. What is the expected amount that you will receive? How much would you be willing to pay to play this game?

4. **St. Petersburg paradox.** You toss independently a fair coin and you count the number of tosses until the first tail appears. If this number is n , you receive 2^n dollars. What is the expected amount that you will receive? How much would you be willing to pay to play this game?

The expected value of the gain for a single game is infinite since if X is your gain, then

$$\mathbf{E}[X] = \sum_{k=1}^{\infty} 2^k \cdot 2^{-k} = \sum_{k=1}^{\infty} 1 = \infty.$$